Energy Reports 12 (2024) 1876–1887

Contents lists available at ScienceDirect

Energy Reports

journal homepage: www.elsevier.com/locate/egyr

Research paper

Modeling carbon dioxide emissions reduction

Andriy Matviychuk [a,1], Olena Zhytkevych [b,*,2], Natalia Osadcha [c,3]

[a] Kyiv National Economic University named after Vadym Hetman, 54/1 Beresteysky Ave., Kyiv 03680, Ukraine
[b] Ukrainian-American Concordia University Office, 1-4, 8/14 Turhenievska Str., Kyiv 01054, Ukraine
[c] Lulea University of Technology, 14 Laboratorievagen, Luleå SE-971 87, Sweden

ARTICLE INFO

JEL Classifications:
C45
C53
Q53
Q56

Keywords:
CO_2 emission target
decarbonization
clustering
self-organizing map

ABSTRACT

The paper proposes a new scientific approach to modeling carbon dioxide (CO_2) emissions in individual countries, based on the tools of Kohonen self-organizing maps and the provisions of economic theory. This work is based on conducted a comprehensive analysis of wide set of indicators influencing carbon dioxide emissions and decarbonization processes in 40 selected countries for 10 years time span from 2013 till 2022. During clustering these nations using self-organizing maps, 14 key indicators were chosen that cover economic and demographic growth, energy consumption, CO_2 emission data, and includes trade of energy recourses for wider analysis and identification of countries with similar decarbonization potential, economic development and trade energy recourses possibilities. This approach exposed distinct clusters of nations with varying decarbonization capabilities and tracked the progress of analyzed countries in terms of CO_2 emission changes in dynamic.

The analysis of clustering was focused on identification leaders and followers in decarbonization activities. The leaders cluster includes Sweden, Norway, New Zealand and other countries exhibited leadership in developing a climate-resilient economy among 40 countries. Hence, all countries positioned closer to the leader cluster on the map demonstrated higher efficiency in their decarbonization pathways. Consequently, these properties of Kohonen maps provide a foundation for formulating general recommendations to achieve an efficient and effective low-carbon economy for followers including Ukraine, Kazakhstan and other energy intensive countries.

Distinct countries decarbonization profiles resulting from clustering were used to develop practical recommendations towards estimating and establishing targeted CO_2 emission levels for analyzed countries, conducting continued promoting renewable energy utilization, enhancing cross-border energy trade, diversifying energy portfolios, etc. Also, the proposed methodology can be utilized to project future emissions trends for each country based on cluster-specific models and facilitate policy decisions making process related to mitigating carbon emissions.

The climate challenge forces countries across the world to engage into decarbonization, however each country defines its own path to reach global targets. This creates additional challenges, since there no generally accepted paths which guarantee efficient outcomes. These transitions rely on the commitment of countries' organizations, firms and individuals to develop and follow steps to move in the mutual direction of decarbonization. One of the first Nationally Determined Contributions (NDCs) defined in 2015 by all 196 countries to the United Nations Climate Convention in Paris (UNDP, 2023). According to the latest NDCs, published in 2020, the modest target of containing climate change to +3°C but achieving them would only represent 30–40 % of what is needed to meet the 2050 2°C goal (EnerData, 2023a).

The UN Sustainable Development Goals (SDGs) is another international agreement (2015) that provides landmarks for the defining paths for countries should follow until 2030 or 2050 (UN Department of Economic and Social Affairs 2023; Zhytkevych, Brochado 2022). These two agreements set long-term targets require a restructuring of economies towards decarbonization and sustainable growing. Hence, all

* Corresponding author.
E-mail addresses: matviychuk@kneu.edu.ua (A. Matviychuk), olena.zhytkevych@uacu.edu.ua, elena.zhitkevich@gmail.com (O. Zhytkevych), nataliia.osadcha@ltu.se (N. Osadcha).
[1] ORCID: 0000-0002-8911-5677
[2] ORCID: 0000-0003-2042-8795
[3] ORCID: 0000-0001-5066-2174

https://doi.org/10.1016/j.egyr.2024.08.004
Received 11 February 2024; Received in revised form 22 June 2024; Accepted 4 August 2024
Available online 9 August 2024

countries faced the main priorities of climate policy and decarbonization activities that would lead to efficient and effective implementation by societies. Furthermore, it requires a careful study of a nation's decarbonization capabilities by using effective economic and mathematics tools.

According to the latest reports (EnerData, 2023b) based on G20 and the rest countries, in 2022 global energy intensity improved (-1.2 %), however it is below 2019–2021 average (-1.9 %/year) and considered to be insufficient to meet the 2°C pathway. Regardless of global economic slowdown, the energy-related CO_2 emissions increase in 2022 (+2.5 %, twice faster than over 2010–2019).

The energy crisis and Russia's invasion of Ukraine foster the EU to foresee further measures having a direct effect on EU greenhouse gas emissions, including new purposes to reduce the EU's dependency on Russian fossil fuels, saving energy and promoting renewable energy (European Scientific Advisory Board on Climate Change 2023).

It is noticeable that climate-neutrality network is expending. As of March 9, 2023, 151 out of 198 countries pledged to achieve carbon neutrality and the rest are in proposed in discussion status. Suriname and Bhutan have already achieved their carbon neutral targets, hence carbon negative countries (Net zero scorecard, 2023).

The countries with legally binding agreements are Sweden, the UK, France, Germany, New Zealand, Finland. There are many countries with in-policy documents including Ukraine, India, China, Kazakhstan and others (Net zero scorecard, 2023).

Hence, the large number of parties of achievement of climate-neutrality requires stronger collaboration among countries in sustainable value chains at national and international levels for increasing resilience and reaching decarbonization targets (Enerdata, 2022; Food and Agriculture Organization of the United Nations 2020; Ministry of Ecology and Natural Resources of Ukraine 2021; Publications Office of the European Union 2019; Turlakova, Lohvinenko 2023).

These together can arise additional challenging obstacles for domestic companies due potential transition risks linked to adjusting regulation, exposure to energy prices, market demand and others.

For effective management of the process of achieving climate-neutrality targets by country, the proactive plan of actions is also required for all players. Taking into consideration the instability of markets conditions, there is no one-fits-all plan for decarbonization at national or international levels. However, this plan should be based on best past experiences and practices of other countries successfully implemented by countries or other authorities with the similar features. Countries with slow decarbonization outcomes and those which are at their first steps towards climate change resilience most likely require additional support and well prepared efficient and effective carbon dioxide reduction program ongoing. Countries with progressive decarbonization economies should continue their ongoing programs and provide support for those that need it for more efficient achievement of global climate-neutrality by 2030 and 2050 results. Consequently, enlargement of mechanisms for defining and monitoring current and potential opportunities for the counties decarbonization path has become relevant. At the same time there are many unsolved aspects of the complex problem primary associated with offsetting carbon footprint at national and consequently at international levels.

The authors tend to show in the paper the competencies of many researchers who emphasize the importance of development and used modern mathematical models to determine the level of CO_2 emissions. Our structured literature review covers the most recent and relevant published papers in the context of key determinants of identification the CO_2 emissions levels of nations and applied methods for modeling national decarbonization potential.

Thus, numerous studies (Cai et al., 2018; Churchill et al., 2018; Jaoua, 2019; Mrabet et al., 2017; Wu et al., 2018; Zhou et al., 2018) have substantiated that CO_2 emissions is a dominant factor in mitigating climate change. The essential issue of energy consumption along with its associated pollution has been addressed in works (Dluhopolskyi et al., 2023; Li et al., 2012; Oliskevych et al., 2018; Schandl et al., 2016). Scholars (Al-mulali et al., 2015; He et al., 2017; Li et al., 2016; Zi et al., 2016) have emphasized the growth in carbon dioxide emissions, attributing it not only to utilization of energy resources but also to growth in trade, urbanization and other related factors. The recent trend of decarbonizing as a potential cost-effective emission reduction is related to the residential buildings, in particularly, the carbon emission mitigation in building operations in large countries like the US, China, India. This has been recognized in the conducted studies (Zhang et al., 2023; Yan et al., 2023) and investigations of the low carbon roadmap and potential carbon neutral pathway of building sector. These findings highlight the complex nature of the challenges created by climate change and relevant factors contributing to this.

Researchers in studying the relationship between agriculture production (Appiah et al., 2018; Sarkodie and Owusu, 2017), energy consumption (Appiah et al., 2019; Dluhopolskyi et al., 2023; Dogan and Seker, 2016; Khvostina et al., 2021; Miroshnychenko et al., 2021), economic growth (Gong et al., 2019; Inekwe et al., 2020; Zomchak et al., 2023), international trade (Babenko et al., 2020; Bielinskyi et al., 2022; Kim et al., 2019; Zharova et al., 2023) and environmental pollution employed diverse approaches. Some scholars (Appiah et al., 2018, 2019; Babenko et al., 2020; Dluhopolskyi et al., 2023; Jaoua, 2019; Kim et al., 2019; Sarkodie and Owusu, 2017; Zomchak et al., 2023) have bended econometric methodologies to quantify the above mentioned relationships, whereas others (Bielinskyi et al., 2022; Khvostina et al., 2021; Miroshnychenko et al., 2021) have employed advanced techniques (e.g. data mining, complexity science, etc.). Hence, we can admit that data science (DS) methods employ comprehensive data-driven approach to support the research in relationships between indirect and factual factors of decarbonization. DS methods systematically demonstrate higher efficiency indicators compared to econometric ones. Literature review allows us to identify that the research and engagement strategies are necessary tools designed to deliver actionable insights to direct efficient low-carbon approaches for nations.

However, taking into account the specifics of the process under study and the periodicity of data on the relevant information resources, it makes sense to model environmental pollution only in annual terms. But in this case, we will get very short time series for each country – just a couple of dozen observations (and if we take into account information on renewable energy sources, then even shorter). And this does not allow the use of modern advanced modeling techniques, which are designed to identify complex patterns in large volumes of data.

Therefore, it becomes advisable to analyze data determining CO_2 emissions for many countries at once. But, given that countries differ very much both in emissions volumes and in the energy, economic, demographic and other indicators that determine them, using one general forecast model for all countries together will be ineffective due to the very high heterogeneity of the data. In this regard, it makes sense to identify more homogeneous groups of countries for subsequent analysis of the dependence of carbon dioxide emissions on many different factors.

To separate such groups of countries, as well as to highlight key factors in datasets, we can apply such a tool in exploratory data analysis as cluster analysis. The field of clustering covers a diverse array of approaches, including the k-methods, minimum spanning tree, hierarchical approaches, and many others (Harkanth and Phulpagar, 2013; Xu and Tian, 2015), proving its flexibility in addressing a wide range of applications.

These clustering methods have their advantages and applications based on their capabilities and functions of practical usages. However, the Kohonen self-organizing map (SOM) toolkit, in addition to the ability to separate study objects, provides a convenient tool for visual analysis of clustering results. The feature of the Kohonen map comparatively to other clustering methods is the ability to directly identify how developed a particular feature is in each studied object compared to others, since the best and worst objects for the set of analyzed indicators are located in opposite sides on the self-organizing map (Kohonen, 2001; Wehrens and Kruisselbrink, 2018). Consequently, the location of the object is important (e.g. the closer the object is to the best angle, the more developed it is in accordance with the studied characteristic). Also, under availability of statistics for long time span, this allows you to study how the analyzed object changes its position on the map over time towards improvement of the studied characteristic. Additionally, analysis of Kohonen maps primary related to the marginal analysis and applied to determine the right actions towards decarbonization path for countries through comparison of nations decarbonization profiles. This makes the Kohonen SOMs more efficient comparatively to other clustering methods in the context of practical usage by analysts.

Clustering successfully used in financial and economic analysis. For example, Hsieh (2005) have used clustering and neural network techniques to design a credit scoring model. Bini and Mathew (2016) have applied this technique in stock data selection then used it as the input for stock price prediction. Babu et al. (2012) have utilized this approach to predict the short-term stock price movements based on the release of financial reports. Turlakova (2022, 2023) used Kohonen SOM for determining the values of the reflexive characteristics of agents in order to ensure effective management of manifestations of herd behavior in an enterprise.

The mentioned works showed that clustering successfully have been utilized in solving various weakly formalized problems and have demonstrated some potentials for addressing problems in other fields, such as identification of CO_2 emission levels. In particular, this possibility has been demonstrated in a number of studies. Gong et al. (2019) applied the clustering method to distinguish provinces with cleaner energy production in China. The authors have concluded that this method also makes it possible to identify specific efforts several provinces have made to implement cleaner energy production practices. However, there are some scope and time limitations to be mentioned: in the paper, the researchers worked with a dataset for one country only and five years plan.

Csereklyei et al. (2021) obtained a model based on clustering approach to examine the energy profiles and trajectories within Australia's National Electricity Market over 10 years period (2011–2019). They have identified 25 distinct electricity generation clusters. The main limitation is that the authors' focus was confined to the energy sector of a single country. This specificity may indeed have implications for the scalability of their approach as it relates only to Australian context.

Inekwe et al. (2020) identified clusters of CO_2 emissions across 72 countries. They have used three key determinants affecting CO_2 emissions (non-renewables, population, and real GDP) and established that in most cases, a two-cluster solution appears to be optimal. However, limiting the number of factors and neglecting the dynamic impact in their analysis may lead to a risk of significant discrepancies in clustering outcomes. Due to unstable consumption levels of renewable and non-renewable energy resources, as well as other input factors over time, the final clustering results can vary significantly.

Recognizing limitations of studies is crucial to addressing them and to interpreting and applying the findings. For example, there are numerous works (Babenko et al., 2020; Csereklyei et al., 2021; Dogan and Seker, 2016; Dupont et al., 2024; Gong et al., 2019; Gyimah et al., 2023; He et al., 2017; Li et al., 2016; Miroshnychenko et al., 2021; Mrabet et al., 2017; Oliskevych et al., 2018; Sarkodie and Owusu, 2017; Véliz et al., 2023; Wu et al., 2018; Zhou et al., 2018; Zi et al., 2016) that keep narrow focus on specific countries or regions like the Chile, China, EU countries, South Africa, US, etc. But keeping the country-specific analysis eliminate the opportunity of generalizing the research results to other countries and effectively adapting the findings and recommendations. Consequently, we have broaden the scope of our study to 40 countries for the ten years period (2013–2022).

Another important finding of our literature review is associated with the identification dominant factors for settling database of clustering. It is obvious that the identification of a country's CO_2 emission target is relevant and continue to develop. The fundamental effect of urban population growth, international trade, economic growth and energy resources consumption on the environment has been realized and should be assessed accurately in order to design policies for mitigating environmental pollution and GHG emissions.

The objective of this study is to determine the decarbonization potential of selected European countries and formulate recommendations for reducing CO_2 emissions by means of advanced mathematical tools.

The method proposed by the authors for analyzing a country's CO_2 emissions by using the Kohonen self-organizing maps consists of the following four stages:

Stage 1. Analysis of recent and the most relevant papers.

We have conducted comprehensive literature review relevant to the creation, evaluation and analysis of countries' CO_2 emission levels and available modelling methodologies.

Outcomes obtained: highlighted key relevant factors for further analysis, covered overview of modelling methods and identified gaps.

Stage 2. Collecting key indicators for the analysis of CO_2 emissions from reputable open sources, checking for gaps, significance, etc. to further create a dataset.

Outcomes obtained: settled the database of key indicators by covering energy consumption, trade, economic growth and demographic conditions of countries.

Stage 3. Carrying out clustering of countries based on the data set from stage 2.

Outcomes obtained: the selected 40 countries have been grouped into several clusters based on key indicators characterizing their CO_2 emission levels.

Stage 4. Analyze self-organizing maps and formulate recommendations for reducing CO_2 emissions for a number of countries.

Outcomes obtained: recommended activities for some countries (e.g. Norway, Sweden and Ukraine) towards forming specific emissions profiles, policy interventions, networks and collaborative initiatives between countries inside and outside their clusters to achieve beneficial improvements.

The proposed systematic approach utilized a comprehensive analysis of countries' CO_2 emissions profiles through clustering methodologies and presented valuable perceptions for policy-making in an environmental management of countries.

The main part of the first stage of the method for analyzing the recent and most relevant papers was performed above in studying the essence of the problem, analyzing previously developed approaches, formulating objective and tasks, and choosing methods for addressing them. Below we will analyze publications in this area in order to form an expanded list of indicators for CO_2 emissions modeling.

Recently studies (Babenko et al., 2020; Miroshnychenko et al., 2021) pointed that one of the vital factors of the energy transition to a low-carbon economy with sustainable electricity sector depends largely on the use of renewables. The renewable usage should be affordable, reliable and sustainable for all nations. However, implementation or increase in usage of the renewable energy sources by developing nations can be challenging due to potential problems of economic, infrastructural, etc. nature. One of the way to eliminate potential challenges is to promote innovation, effective usage of the energy resources and natural resources and enhance comparative advantage to facilitate trade.

For example, effective usage of energy storage, network expansion, smart grids, new business models and market arrangements (Dupont et al., 2024; Gyimah et al., 2023; Miroshnychenko et al., 2021; Véliz et al., 2023) can lead to effective designing or development the renewable energy strategy by nations. Taking all these factors into consideration, we decided to extend the set of key indicators to trade possibilities of energy recourses between countries. This helped us to understand comparative and absolute advantages of countries in terms of energy recourses excess and shortages.

Another critical factors for developing countries are the cost of electricity due to the existence of restrained demand (Relva et al., 2021). So, these countries have to consider about increase energy supply. The main outcomes from studies about the role of the trade-offs and synergies between climate change and energy security, climate change and universal electricity access and energy sufficiency and mobility have been contributed to the set of inputs for our further study and analysis.

Identifying the key factors that affecting countries' CO_2 emissions is vital to initiate the process of clustering nations based on their decarbonization capabilities. Such identification requires analysis of numerical statistical data collected over a sufficiently long period (for example, 10 years). The analysis of databases have been described in our previous studies (Zhytkevych et al., 2022, 2023). Also, we updated to 2022 year and expanded them with new indicators.

The length of the time period of 10 years is coincides with starting active decarbonization efforts of many nations, mostly after Paris Agreement of 2015. Making a longer interval is not relevant since it can lead to confused conclusions that take into account outdated patterns. Hence, this study is not a time series forecast and proposed model is trained on the data of one cluster. So, if 10 countries fall into it, then 100 records are already enough to obtain meaningful conclusions, especially if we take into account consideration of more than 10 factors (14 in our study).

Thus, the input data set for the clustering analysis was obtained from the "World Development Indicators" database provided by the World Bank (2022) and the Enerdata's "World Energy & Climate Statistics – Yearbook 2023" (EnerData, 2023c). The study ensures transparency and accessibility of information by utilizing information from open databases. Also it allows comprehensive examination of various factors affecting CO_2 emissions across different countries. Additionally, the open data facilitates reproducibility and enables other researchers to validate and build upon the findings of the study.

Within the stage 2, we have collected data with further indicator validation for a dataset creation. Overall, the integration of appropriate analysis of numerical statistical data obtained from open sources plays a crucial role in forming a strong input data set. Ultimately, this contributes to the consequential clustering of countries based on their influencing factors related to decarbonization.

In our study, number of analyzed countries is 40 and they have been selected due to fulfillment and completeness criteria (e.g. key energy and climate features of nations).

Then we justified the sets and subsets of indicators (characteristics) which affect decarbonization potential of countries and approached factors influencing CO_2 emissions across analyzed countries (Zhytkevych et al., 2022, 2023).

The procedure of selecting key indicators combined analysis of economic, social, energy, climate and trade of natural recourses aspects of nations. This process complies with conditions of completeness, effectiveness and efficiency criterion and avoids double counting of indicators.

Our study follow the tendency of recent research (Appiah et al., 2019; Gong et al., 2019; Zhou et al., 2018; Zhytkevych et al., 2022, 2023; Zi et al., 2016; Zomchak et al., 2023) on the energy, environmental linkage with expanding renewable energy components, economic growth and urbanization. However, this study attempts to disclose the influence of nations' energy recourses consumption (excluding their production), economic growth rate and current CO_2 emissions on decarbonization capabilities of countries. In this paper, authors proposed to add new indicators associated with trade of energy recourses (e.g. electricity, natural gas, oil products, coal and lignite) to highlight the cross-border integration between nations. Such synergy between countries is important for decarbonization effectiveness and makes collaboration towards decarbonization at national and international levels more efficient.

Indicators gaps in the database have been substituted with average values for respective groups of countries they correspond to (e.g. EU, G7, OECD, CIS, etc.) and given in the reports (EnerData, 2023c; The World Bank, 2022). To reduce the excessive influence of variables with large absolute values, min-max normalization was performed (normalized value of 0 corresponds to the smallest value of the appropriate indicator and 1 to the largest for the period of 2013–2022). Also, the new three factors has been added to the updated set of indicators in comparison with researches (Zhytkevych et al., 2022, 2023). The Table 1 composes data set from two reports (EnerData, 2023c; The World Bank, 2022) and represents normalized values of 14 indicators in the range from 0 to 1 for 40 countries for 2022. Moreover, the proposed list of indicators can be reduced in the process of conducting an experimental study. This happens if a certain indicator does not have a significant impact on the clustering process (e.g. its values are evenly distributed across different clusters). In our analysis all indicators become significant.

In the stage 3 we conducted and presented results of clustering nations by using clustering methodology of Kohonen self-organizing maps (Kohonen, 2001). Thereby, countries were classified according to their potential capacity for CO2 emission reduction on the basis of indices presented in Table 1. The self-organizing map framework has been utilized because it can generate homogeneous groups of countries and provides a convenient tool for visual analytical study of the results obtained.

The outcome of constructing the Kohonen map yields a visual depiction of a two-dimensional grid of neurons, which mirrors the organizational arrangement of the studied countries. This grid forms clusters of countries that exhibit similarity in terms of the gathered indicators, encompassing factors like economic growth, energy consumption, CO_2 emission levels, and so forth (Fig. 1).

The clustering of 400 data vectors (40 countries over 10 years) with relevant 14 key indicators from the consolidated database (see Table 1) was automated using Deductor Studio Academic software package.

The map construction process is related to determining its optimal dimension of the number of neurons, which was selected from various options based on the weighted average quantization error criterion. It determines the average distance between the data vector supplied into the map inputs and the neurons parameters.

The SOM constructions were conducted by utilizing a hexagonal grid with dimensions of 16 by 12 neurons. The clustering process has been executed over 5500 learning epochs for high-quality visualization and ease of analysis. The Gaussian function has been chosen for determining the neighborhood of neurons for the cooperation procedure.

The most suitable number of clusters for representing the self-organizing map of analyzed countries has been selected through several experiments and represented on Figs. 2 and 3.

The 40 countries analyzed were grouped into three (Fig. 2) and six clusters (Fig. 3), numbered 0–2 and 0–5, respectively (the cluster number corresponds to the color on the ruler at the bottom of the Kohonen map of each figure).

At stage 4 of the method, self-organizing maps were considered and recommendations were generated based on the content of the resulting clusters.

The initial series of clustering procedures utilizing 14 indicators for 40 countries over the period of 2013–2022 was completed with three clusters (see Fig. 2). The examination of the composition of these clusters showed that certain countries were grouped together based on similar only 9 among given 14 indicators mostly related to energy consumption and emissions, however exhibited differing values in terms of economic growth factors. For instance, one of the three clusters included Ukraine, Italy, Canada, etc. It is important to mention that such differences could lead to potential obstructers and make further clustering analysis uncertain. One of the main objectives of research is to identify countries with similar levels across the vast majority of selected indicators, rather than just a few. Consequently, to address this issue, we've decided on increasing the number of clusters to change their composition. Upon conducting several iterations, it was determined that increasing the number of clusters from 3 to 6 proved an effective

Table 1

Database of 14 indicators for defining decarbonization profiles of 40 nations for 2022.

Countries	Year	Total energy consumption	Energy intensity of GDP	Coal and lignite trade	Coal and lignite consumption	Oil products balance of trade	Oil products consumption	Natural gas balance of trade	Natural gas domestic consumption	Electricity balance of trade	Electricity domestic consumption	Share of renewables in electricity	Average CO2 emission factor	GDP per capita growth (annual %)	Urban population (% of total population)
Algeria	2022	0.01	0.46	0.58	0	0.67	0.02	0.49	0.05	0.56	0.02	0.58	0.59	0.58	0.85
Argentina	2022	0.02	0.27	0.58	0	0.78	0.04	0.64	0.05	0.62	0.03	0.69	0.58	0.59	0.91
Australia	2022	0.03	0.30	0.13	0.02	0.93	0.06	0.36	0.05	0.58	0.05	0.69	0.59	0.59	0.89
Belgium	2022	0.01	0.23	0.59	0	0.77	0.02	0.67	0.02	0.55	0.03	0.67	0.58	0.59	0.94
Brazil	2022	0.08	0.28	0.60	0	0.84	0.14	0.65	0.03	0.61	0.08	0.90	0.58	0.59	0.90
Canada	2022	0.07	0.70	0.54	0	0.71	0.12	0.47	0.15	0.39	0.08	0.83	0.58	0.58	0.88
Chile	2022	0	0.19	0.59	0	0.82	0.02	0.64	0.01	0.58	0.03	0.78	0.58	0.58	0.90
China	2022	1	0.56	0.96	1	0.78	0.86	0.96	0.39	0.51	0.59	0.69	0.59	0.59	0.81
Colombia	2022	0.01	0.05	0.50	0	0.78	0.02	0.63	0.01	0.58	0.03	0.85	0.58	0.60	0.88
Czech Republic	2022	0.01	0.32	0.58	0.01	0.78	0.01	0.65	0.01	0.53	0.03	0.63	0.58	0.58	0.85
Egypt	2022	0.02	0.09	0.59	0	0.78	0.04	0.62	0.07	0.57	0.04	0.62	0.58	0.59	0.73
France	2022	0.05	0.13	0.59	0	0.91	0.08	0.73	0.04	0.63	0.07	0.67	0.58	0.59	0.87
Germany	2022	0.07	0.09	0.63	0.04	0.81	0.12	0.83	0.09	0.48	0.08	0.74	0.59	0.58	0.86
India	2022	0.26	0.30	0.89	0.25	0.67	0.3	0.69	0.07	0.56	0.13	0.66	0.59	0.60	0.71
Indonesia	2022	0.07	0.18	0	0.05	0.89	0.09	0.58	0.04	0.58	0.04	0.64	0.59	0.59	0.79
Italy	2022	0.03	0.07	0.60	0	0.71	0.06	0.8	0.07	0.73	0.06	0.71	0.58	0.59	0.84
Japan	2022	0.10	0.15	0.81	0.04	0.88	0.19	0.86	0.10	0.58	0.14	0.66	0.59	0.58	0.91
Kazakhstan	2022	0.01	0.56	0.54	0.02	0.76	0.01	0.61	0.02	0.58	0.03	0.62	0.59	0.58	0.79
Malaysia	2022	0.02	0.32	0.62	0.01	0.80	0.04	0.55	0.05	0.57	0.03	0.64	0.59	0.60	0.86
Mexico	2022	0.04	0.17	0.60	0	0.96	0.12	0.78	0.08	0.58	0.05	0.66	0.59	0.59	0.87
Netherlands	2022	0.01	0.08	0.59	0	0.68	0.02	0.68	0.04	0.56	0.03	0.72	0.58	0.59	0.92
New Zealand	2022	0	0.29	0.58	0	0.80	0	0.47	0	0.58	0.02	0.89	0.58	0.58	0.89
Nigeria	2022	0.04	0.63	0.58	0	0.88	0.02	0.58	0.02	0.58	0.02	0.68	0.58	0.58	0.77
Norway	2022	0	0.17	0.58	0	0.77	0.01	0.32	0.01	0.53	0.03	0.94	0.58	0.59	0.88
Poland	2022	0.02	0.17	0.59	0.02	0.79	0.04	0.66	0.02	0.57	0.04	0.65	0.59	0.60	0.80
Portugal	2022	0	0.07	0.58	0	0.77	0.01	0.64	0.01	0.61	0.02	0.80	0.58	0.60	0.82
Romania	2022	0	0.04	0.58	0	0.76	0.01	0.63	0.01	0.58	0.02	0.73	0.58	0.60	0.78
Russia	2022	0.21	0.99	0.36	0.05	0.22	0.19	0.23	0.57	0.54	0.12	0.64	0.58	0.57	0.85
Saudi Arabia	2022	0.06	0.57	0.58	0	0.48	0.17	0.53	0.11	0.58	0.05	0.58	0.59	0.60	0.89
South Africa	2022	0.03	0.62	0.49	0.03	0.85	0.02	0.64	0.01	0.57	0.04	0.61	0.59	0.58	0.83
South Korea	2022	0.07	0.46	0.74	0.02	0.66	0.13	0.78	0.06	0.58	0.08	0.61	0.58	0.59	0.87
Spain	2022	0.03	0.09	0.59	0	0.76	0.06	0.71	0.04	0.50	0.05	0.73	0.58	0.59	0.87
Sweden	2022	0.01	0.20	0.58	0	0.73	0.01	0.63	0	0.46	0.03	0.83	0.58	0.58	0.90
Thailand	2022	0.03	0.32	0.61	0.01	0.76	0.07	0.67	0.04	0.70	0.04	0.64	0.58	0.59	0.77
Turkey	2022	0.04	0.03	0.63	0.03	0.83	0.05	0.76	0.06	0.59	0.04	0.73	0.59	0.59	0.86
Ukraine	2022	0.01	0.84	0.59	0.01	0.81	0.01	0.64	0.02	0.57	0.04	0.64	0.58	0.51	0.83
United Arab Emirates	2022	0.02	0.44	0.59	0	0.70	0.03	0.66	0.07	0.58	0.03	0.59	0.58	0.60	0.9
United Kingdom	2022	0.04	0	0.59	0	0.81	0.06	0.71	0.08	0.56	0.06	0.73	0.58	0.59	0.88
United States	2022	0.57	0.32	0.49	0.10	0	0.95	0.34	1	0.73	0.49	0.66	0.58	0.58	0.88
Uzbekistan	2022	0.01	0.63	0.58	0	0.77	0	0.61	0.05	0.59	0.02	0.61	0.59	0.59	0.76

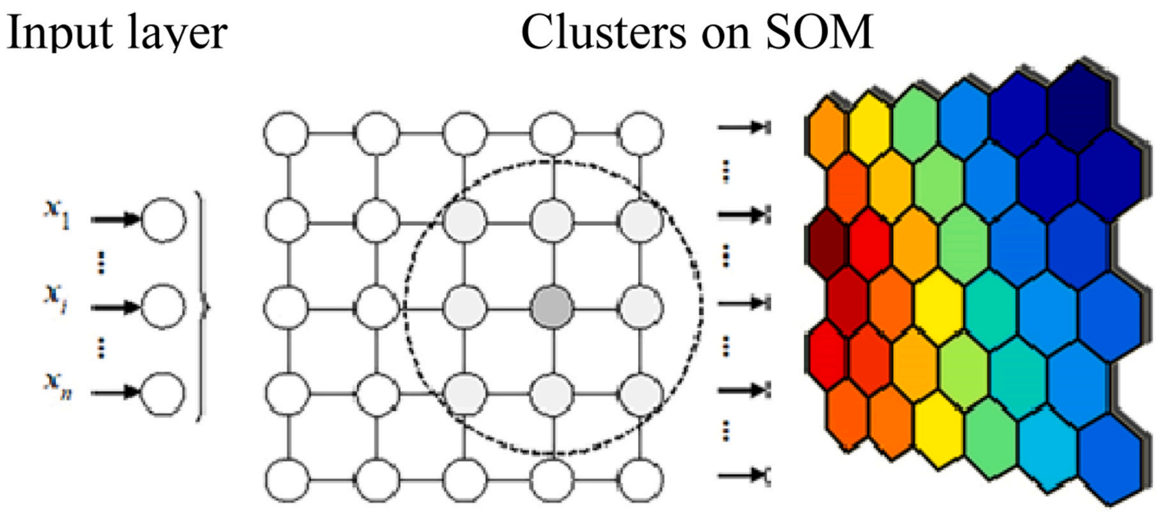

Fig. 1. Visual illustration of clusters on Kohonen map (Matviychuk et al., 2019).

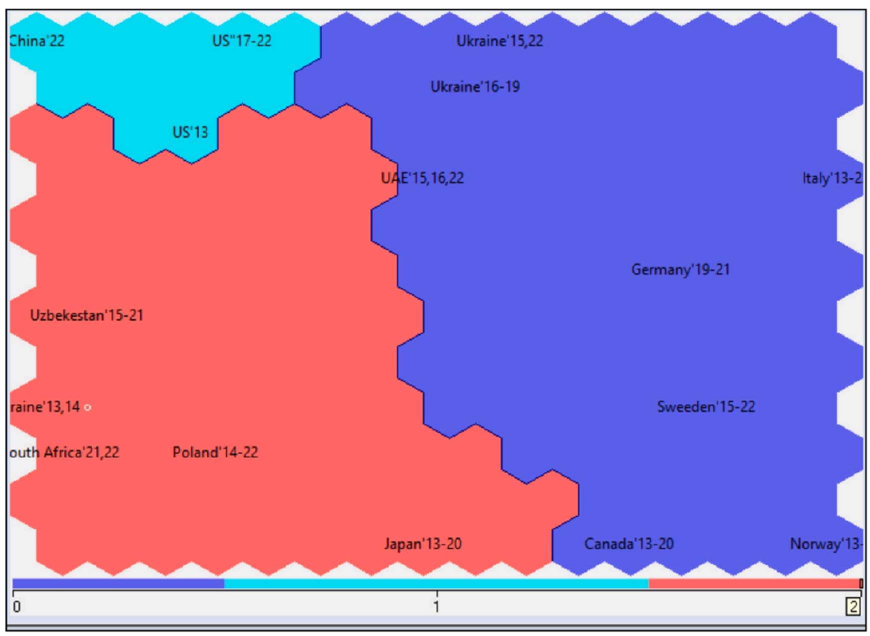

Fig. 2. Kohonen map of three clusters from 40 countries based on 14 indicators for 2013–2022.

correction. Thereby, many countries were grouped together based on similar characteristics, including decarbonization levels and economic growth (see Fig. 3).

The further increasing number of clusters to ten and etc. was unsuccessful, since some countries with distinct profiles by decarbonization activities (e.g., Canada, China, Norway, US) composed single cluster, hence the main idea of clustering was neglected.

So, the Kohonen map of six clusters from 40 countries based on 14 indicators for ten years span of 2013–2022 (see Fig. 3) was proposed to be used for the representing results, analysis and developing recommendations in the paper. The summarized set of clusters for analyzed countries is represented in Table 2.

The contents of the Table 2 illustrates the dynamic changes in the position of countries on the map as a decisive factor for further analysis of the success of decarbonization activities. For example, there are some countries (e.g., Chile, Saudi Arabia, South Korea, UK, Ukraine, US, etc.) performed multiple positional shifts over the span of 10 years. The Kohonen map facilitated the tracking and analysis of these movements. For instance, Chile transitioned from cluster 0–4 then returned back to cluster 0, while the United Kingdom and Mexico transferred from cluster 4–0. These movements within and between clusters can be used for analysis and tracking the improving or deteriorating countries' decarbonization positions relatively to others or itself within 10 years period.

Such benchmarking requires attributing "leaders" and "followers" in decarbonization activities for 2013–2022. According to the clustering results for 40 countries composed in clusters 0–5, there are some countries with outstanding energy, economic growth and CO_2 emission indicators, thus these nations could be characterized as low-carbon economies. These countries referred to cluster 1 (see Fig. 3). Therefore, this cluster has been labeled "leader" and countries inside the clusters are leaders while the rest – followers. For that reason, if countries were shifted closer to the leader cluster 1, they improved their decarbonization positions, vice versa was true. So, the UK and Mexico have enhanced their positions, since they transferred to cluster 0, which is closer to the border of the cluster 1. Moreover, the degree of improvement could be approached in this way – the closer to the border

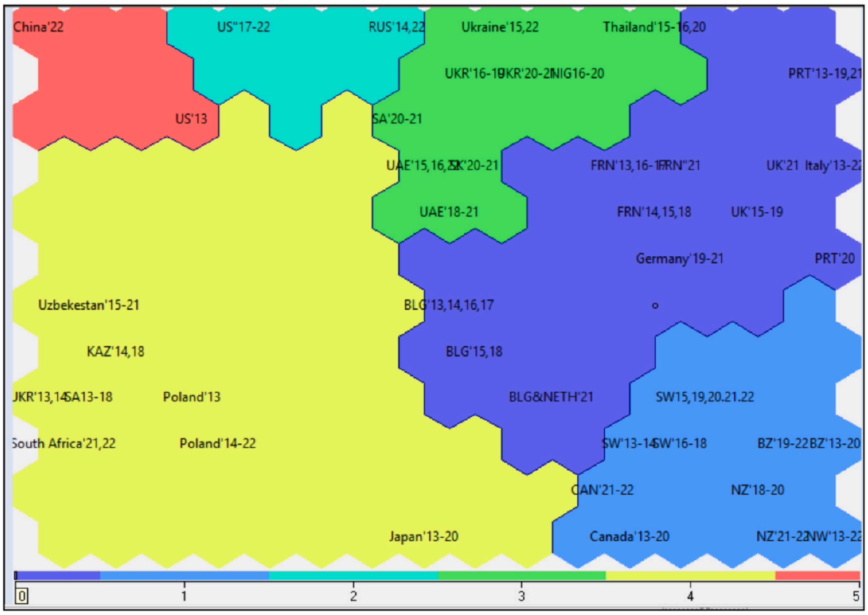

Fig. 3. The Kohonen map of six clusters from 40 countries based on 14 indicators for 2013–2022.

Table 2
Clusters of the self-organizing map based on 14 indicators for 40 countries in the period 2013–2022.

Cluster 0	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5
Argentina (2013–2022) Belgium (2013–2022) Chile (2013, 2014, 2016–2020, 2022) Czech Republic (2020–2022) Egypt (2019–2020, 2022) France (2013–2022) Germany (2019–2022) Indonesia (2013) Italy (2013–2022) Mexico (2020–2022) Netherlands (2019–2022) Romania (2013–2022) Portugal (2013–2022) Spain (2013–2022) United Kingdom (2015–2022)	Brazil (2013–2022) Canada (2013–2022) Colombia (2013–2022) New Zealand (2013–2022) Norway (2013–2022) Sweden (2013–2022)	Russia (2013–2022) US (2017–2022)	Nigeria (2013–2022) Saudi Arabia (2020–2021) South Korea (2020–2022) Thailand (2013–2022) UAE (2015–2016, 2018–2022) Ukraine (2015–2022) Uzbekistan (2018)	Algeria (2013–2022) Australia (2013–2022) Chile (2015, 2021) Czech Republic (2013–2019) Egypt (2013–2018, 2021) Germany (2013–2018, 2022) India (2013–2022) Indonesia (2014–2022) Japan (2013–2022) Kazakhstan (2013–2022) Malaysia (2013–2022) Mexico (2013–2019) Netherlands (2013–2018) Poland (2013–2022) Saudi Arabia (2013–2019, 2022) South Africa (2013–2022) South Korea (2013–2019) Turkey (2013–2022) United Arab Emirates (2013–2014, 2017) Ukraine (2013–2014) United Kingdom (2013–2014) Uzbekistan (2013–2017, 2019–2022)	China (2013–2022) US (2013–2016)

of leader cluster, the greater improvement toward decarbonization.

Concerning leader's cluster, which is composed by Brazil, Canada, Colombia, New Zealand, Norway and Sweden, it showed a high level of electrification and urbanization along with an average growth rate in GDP per capita for the analyzed period from 2013 to 2022. Furthermore, these countries demonstrated a low level of domestic consumption of coal and lignite, coupled with an average CO_2 emission factor (Fig. 4). Hence, these countries (cluster 1) show good examples in reducing CO_2 emissions and deserve to be called low-carbonized economies among the 40 countries analyzed for 2013–2022.

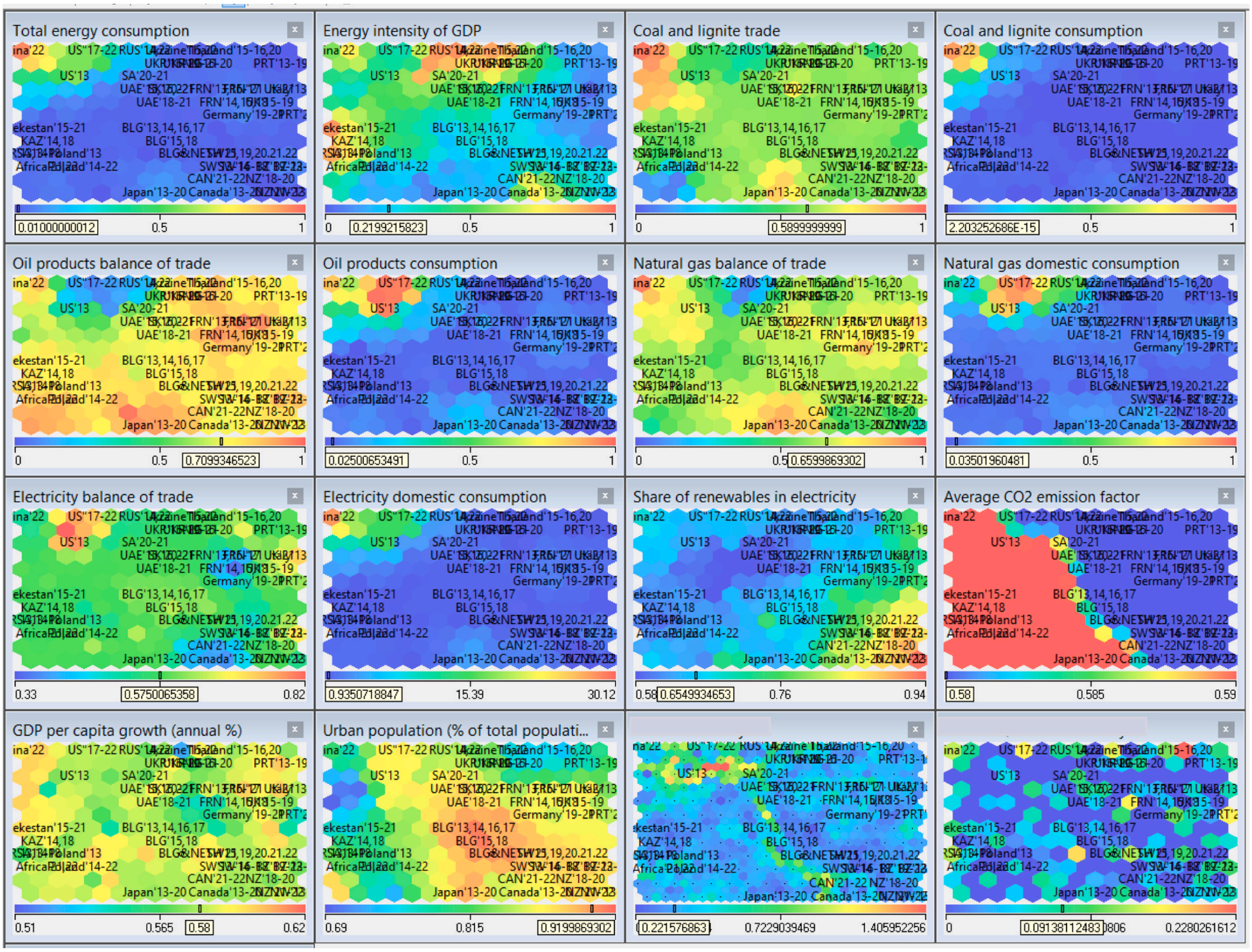

Fig. 4. The heatmaps of 14 indicators on the data of 40 countries for 2013–2022.

Note that Sweden (2013–2022), Norway (2013–2022) and New Zealand (2013–2022) exhibited low levels of energy intensity of GDP, as well as reduced consumption of oil products, natural gas, electricity and showed the highest proportions of renewable, wind, and solar energy in electricity generation among the analyzed countries (Fig. 4). These gave a strong argument that countries from cluster 1 at the lower right corner of the SOM in Fig. 3 are positioned as the most environmentally advanced among 40 countries analyzed for the last 10 years.

As we mentioned before, the closer country to the leader cluster, the greater improvement toward decarbonization. Thus, cluster 5 exhibits the least favorable values across the 14 indicators compared to the remaining five clusters, and it is located opposite the leader cluster on the map (upper left corner in Fig. 3).

The countries of the cluster 5 (see Table 2) that showed sluggish progress in their decarbonization efforts are China (2013–2022) and the US (2013–2016). In 2017, the US changed its position from cluster 5 to cluster 2 and stayed there till 2022. This movement associated with some improvement in decarbonization activities, since cluster 2 is closer to cluster 1 than cluster 5.

Countries like Belgium, France, Germany, Italy, Portugal consistently stayed in the cluster 0 for the entire 10-year period, however they moved within the cluster towards the cluster 1's border. These indicated some improvements in decarbonization activities (Fig. 5).

Considering the Ukrainian decarbonization case through analysis of the SOM at Fig. 5, it was observed that Ukraine changed its positions several times with different degrees of improvement for 2013–2022. For example, it initially positioned within cluster 4 for 2013–2014, composed by countries like Czech Republic (2013–2019), Germany (2013–2018, 2022), Kazakhstan (2013–2022), Poland (2013–2022) and others (see Table 2). During that period, Ukraine and the mentioned above countries exhibited notably high energy intensity and urbanization with low levels of domestic consumption for oil products and electricity. Additionally, it is noticeable that there were limited contributions from renewable sources, solar, and wind energy in electricity generation for these countries (see Fig. 4).

Upon further analysis, it was observed that Ukraine (2013–2022) exhibited an average level of electrification, urban population, CO_2 emission factor, and a growth rate in GDP per capita. Ukraine demonstrated a relatively low level of domestic consumption for coal and lignite, high coal and lignite trade, oil products balance of trade and natural gas balance of trade coupled with electricity balance of trade (Fig. 4).

Examining the arrangement of Ukraine in 2022, we have to admit the adverse position associated with the shrinking improvements reached from 2016 to 2021. Initially, Ukraine was assigned to cluster 4 in 2013–2014, then transitioned to cluster 3 and stayed there until 2022. However, it is visible, that it moved closer to the cluster 2 border in 2015, then in 2016 moved to the middle position of the cluster 3 and stayed there till 2021. These improvements could be explained by particular series of events described by analytical reports from international organizations and public authorities (Food and Agriculture Organization of the United Nations 2020; Ministry of Ecology and

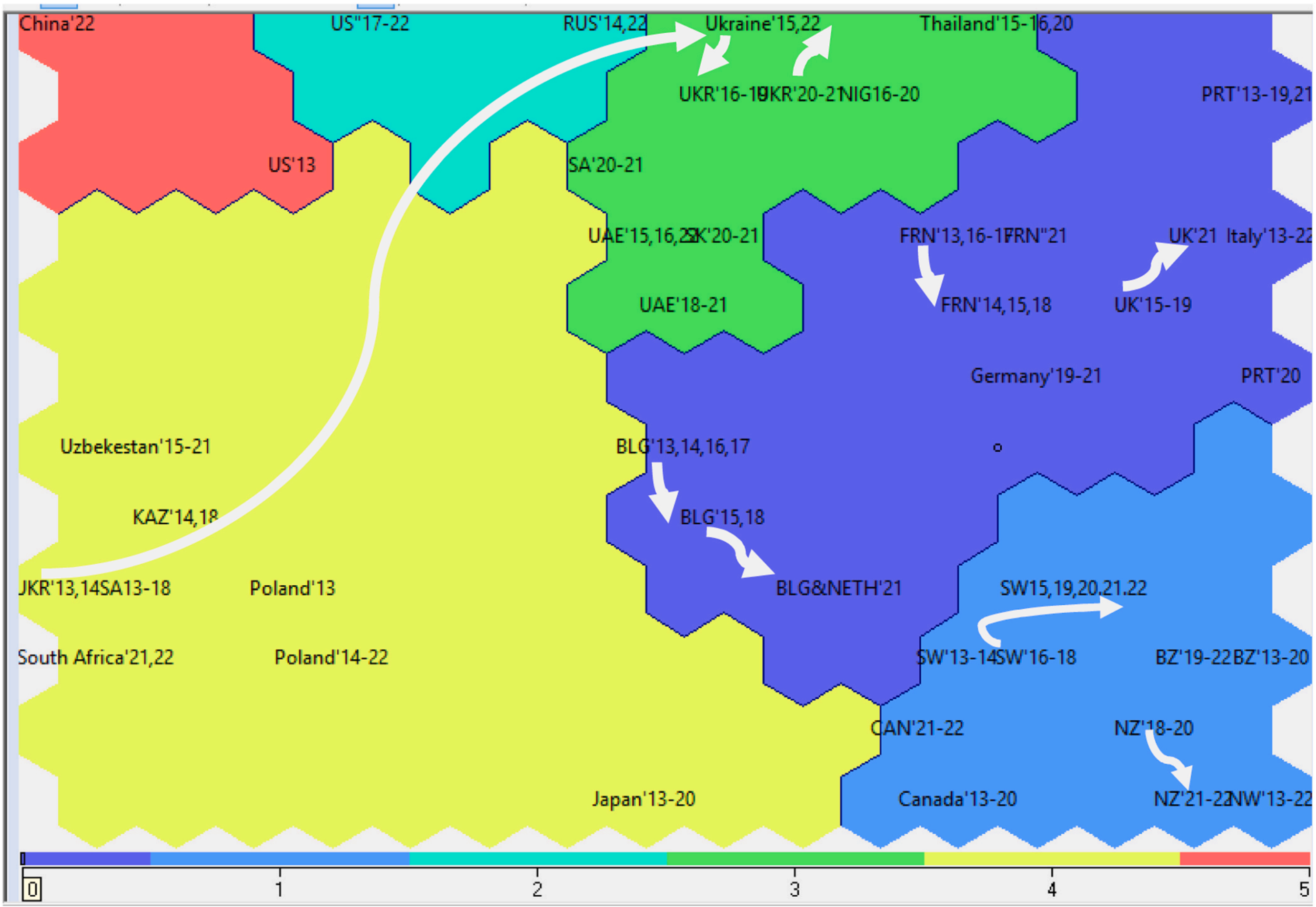

Fig. 5. The trajectory of movement of some countries between clusters on the Kohonen map for 2013–2022.

Natural Resources of Ukraine, 2021; UNICEF, 2021). Since then, Ukraine contended with significant territorial shifts such as temporary annexation of the Autonomous Republic of Crimea and Sevastopol city by the Russian Federation along with the anti-terrorist operations in individual districts of Donetsk and Lugansk regions from 2014 (Food and Agriculture Organization of the United Nations 2020). These events had a significant impact on Ukraine's development trajectory on the SOM.

However, Ukraine made notable improvements in the development of renewable energy in the period of 2015–2020. The share of renewable energy sources in electricity production surged from 7.9 % to 11.3 %. This period was associated with substantial reductions in energy consumption, greenhouse gas emissions, pollutants and created some good conditions for the implementation of various economic enhancements (Ministry of Ecology and Natural Resources of Ukraine, 2021; UNICEF, 2021).

In 2015, the Ministry of Ecology and Natural Resources of Ukraine initiated the development of the national strategy on the alignment of Ukrainian legislation with EU legislation for environmental protection. Significant four-fold increase in the tax on greenhouse gas emissions occurred in 2019.

In 2021, the government introduced a new directive that the large and medium-sized industrial enterprises must formulate plans for monitoring greenhouse gas emissions. In July 2020, Ukraine officially authorized the European green deal, a comprehensive initiative aimed to achieve the climate neutrality across the European continent by the year 2050. In March 2021, the Cabinet of Ministers of Ukraine ratified the National Economic Strategy for the period until 2030, outlined the pathway to achieve climate neutrality by 2060 and other events with potentially favorable impacts on decarbonization position of the country (Ministry of Ecology and Natural Resources of Ukraine, 2021; UNICEF, 2021).

However, the situation was changed in February 2022 and Ukraine returned back to its position in 2015 (see Fig. 5). Russia's invasion of Ukraine had direct impact on the Ukrainian and the EU greenhouse gas emissions, but at the same time could foster the process to reduce the EU's dependency on Russian fossil fuels, saving energy and promoting renewable energy (European Scientific Advisory Board on Climate Change, 2023).

The countries movements between and within clusters happened due to changing and shifting in their key indicators like energy product consumption and implementing policies towards reducing CO_2 emissions. From the economics perspective, these movements reflect changes in a country's status from being high-emission to low-emission economies and vice versa. Therefore, the analysis of SOM pointed two main findings: effective managing natural recourses and spaces usage as well as setting up and maintaining trade conditions are important factors; and the cross-countries collaboration for a low-carbon cooperation between countries could be initiated and enhanced. Based on the above, decision makers (policy makers, advisors, etc.) of countries with high energy intensities should use their comparative advantages in trade and managing their energy recourses for improving the overall energy consumption and trade conditions to benefit all parties.

The SOMs provide useful information to identify decarbonization capabilities of countries, also it could be used to search the optimal position for a country on the map for fostering its resilient economy. To do so technically, countries from the follower clusters should align their 14 indicators with the values of the leader indicators.

The findings from this analysis can be represented as a valuable benchmarking tool for countries seeking to initiate or enhance their decarbonization efforts. Nations (followers) via studying the practices of leading countries can adopt and adapt "best practices" for more efficient results. Hence, the explicit assessment of the country's effectiveness in low-carbon activities can be conducted by comparing a country's decarbonization profile against leaders' profiles driving the country's movements towards carbon neutrality associated with the establishment of a specific target for carbon dioxide emissions aligning with purposes of a resilient and efficient economy. However, one of the way to reach this global efficiency, all countries should enhance the cooperation with one another with their best potentials. This approach enables to leverage historical examples from active economies in low-carbonizing by countries with sluggish decarbonization activities. Thereby, this course is associated with minimizing potential difficulties in the decarbonization process at national and international levels.

Based on outcomes of our analysis and taking into account 40 analyzed nations' economic, energy and CO_2 emission potentials, trade balance of energy resources, we believed that cooperation between countries inside and outside the formed clusters could bring effective and efficient outcomes for the global decarbonization. The authors formed some recommendations (Table 3) towards fostering efficient and effective low-carbon economies by leaders (several countries from cluster 1) and followers (few countries from other clusters).

As can be seen from the study, clustering is a powerful analytical tool which assists analysts in forming distinct countries decarbonization profiles, categorizing countries as either high or low CO_2 emitters. Fatherly, this information can be used for estimating and establishing targeted CO_2 emission levels for analyzed countries. The proposed methodology can be utilized to project future emissions trends and facilitate policy decisions making process related to mitigating carbon emissions.

1. Conclusions

The development of effective mathematical models contribute the accurate determining CO_2 emission levels and implementing effective and efficient decarbonization activities. The authors have proposed the scientific approach to achieve this objective, which, based on Kohonen self-organizing maps, makes it possible to segment countries over time in the field of decarbonization, allowing the identification of leaders that can be an example for other countries to follow.

Given the properties of applied clustering method, the closer a country is positioned to the leader cluster on the map, the more advanced and efficient it is in terms of decarbonization activities and vise-versa is true. This deduced a roadmap and can guide countries, which are in process of developing their decarbonization efforts towards effective low-carbon economies.

The analysis of created maps illustrated progress of steadily advanced countries like Sweden, Norway and New Zealand within a leader cluster over ten years 2013–2022, while Ukraine showed variable dynamics in its decarbonizing pathway. Following the outcomes of comprehensive analysis of SOM, the recommendations customized to Ukraine and other follower countries, as well as leader countries have been formulated. These guidance are aimed to facilitate the most effective transition towards low-carbonization processes at national and international levels by all participants.

The main outcomes of the four-stages approach can be outlined as: covered overview of modelling methods and identified gaps and advantages for set problem; settled the database of key indicators by covering energy consumption, trade, economic growth and demographic conditions of countries; the selected 40 countries have been grouped into several clusters based on key indicators characterizing their CO_2 emission levels and efforts to decarbonize the economy; recommended activities for some countries towards forming specific emissions profiles, policy interventions, networks and collaborative initiatives between countries inside and outside their clusters to achieve beneficial improvements.

Thus, the paper exhibits the application of a scientific method for identification countries decarbonization profile, hence setting a country's direction towards efficient carbon mitigation. The findings presented in this study provide significant implications for academia and policymakers and also contributes to the future research endeavors.

Author Statement

The answers for reviewers' comments have been provided by authors. We have revised our work accordantly and provided answers in the Revised manuscript with track changes.

CRediT authorship contribution statement

Olena Zhytkevych: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Andriy Matviychuk:** Writing – review & editing, Writing – original draft, Supervision, Project administration,

Table 3
Recommendations towards low-decarbonization economies based on clustering results.

Cluster: Leader
Selected countries: Sweden, Norway and New Zealand
Conduct continued promoting renewable energy utilization
continue prioritizing renewable energy sources;
provide sustain efforts towards decreasing consumption of fossil fuels (e.g. oil products, natural gas, coal and lignite);
fostering the integration of renewables (e.g. wind, and solar energy) in electricity production.
Enhance cross-border energy trade
control and leverage surplus energy resources (e.g. renewable) for cross-border trade;
use the proceeds from trade of energy surplus for further improvement of the renewable energy utilization;
promote energy security and economic cooperation with energy recourse shortage countries.
Conduct technology sharing and innovation
assist and promote the exchange of renewable energy technologies and best practices with countries to enhance the low-carbon energy systems and therefore spread positive externalities.
Clusters: Followers
Selected countries: Ukraine, Kazakhstan, Uzbekistan and other carbon intensive countries
Diversify energy portfolio
find potentials for renewable energy development and reduce dependency on fossil fuels;
focus on decreasing indicators related to oil products, natural gas, coal, and lignite domestic consumption and simultaneously increasing the share of renewables (e.g. wind, and solar in electricity consumption).
Promote and actively participate in collaborative research and development
engage in common research and development projects with EU countries to advance renewable energy technologies and network integration solutions;
align national policies and targets with EU directives related to renewable energy integration and emission reduction activities.

Methodology, Conceptualization. **Natalia Osadcha:** Project administration, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

References

Al-mulali, U., Weng-Wai, C., Sheau-Ting, L., Mohammed, A.H., 2015. Investigating the environmental Kuznets curve (EKC) hypothesis by utilizing the ecological footprint as an indicator of environmental degradation. Ecol. Indic. 48, 315–323. https://doi.org/10.1016/j.ecolind.2014.08.029.

Appiah, K., Du, J., Poku, J., 2018. Causal relationship between agricultural production and carbon dioxide emissions in selected emerging economies. Environ. Sci. Pollut. Res. 25, 24764–24777. https://doi.org/10.1007/s11356-018-2523-z.

Appiah, K., Du, J., Yeboah, M., Appiah, R., 2019. Causal Correlation between energy use and carbon emissions in selected emerging economies—panel model approach. Environ. Sci. Pollut. Res. 26, 7896–7912. https://doi.org/10.1007/s11356-019-04140-2.

Babenko, V., Rayevnyeva, O., Zherlitsyn, D., Dovgal, O., Goncharenko, N., Miroshnichenko, T., 2020. Dynamics of forecasting the development of renewable energy technologies in Ukraine and Chile. Int. J. Ind. Eng. Prod. Res. 31 (4), 587–596. https://doi.org/10.22068/ijiepr.31.4.587.

Babu, M., Geethanjali, N., Satyanarayana, B., 2012. Clustering approach to stock market prediction. Int. J. Adv. Netw. Appl. 3 (4), 1281–1291. ⟨https://www.ijana.in/papers/V3I4-10.pdf⟩.

Bielinskyi, A.O., Matviychuk, A.V., Serdyuk, O.A., Semerikov, S.O., Solovieva, V.V., Soloviev, V.N., 2022. Correlational and non-extensive nature of carbon dioxide pricing market. In: Ignatenko, O., Kharchenko, V., Kobets, V., et al. (Eds.), Communications in Computer and Information Science: Vol. 1635. ICTERI 2021 Workshops. Springer, Cham, pp. 183–199. https://doi.org/10.1007/978-3-031-14841-5_12.

Bini, B.S., Mathew, T., 2016. Clustering and regression techniques for stock prediction. Procedia Technol. 24, 1248–1255. https://doi.org/10.1016/j.protcy.2016.05.104.

Cai, Y., Sam, C.Y., Chang, T., 2018. Nexus between clean energy consumption, economic growth and CO2 emissions. J. Clean. Prod. 182, 1001–1011. https://doi.org/10.1016/j.jclepro.2018.02.035.

Churchill, S.A., Inekwe, J., Ivanovski, K., 2018. Conditional convergence in per capita carbon emissions since 1900. Appl. Energy 228, 916–927. https://doi.org/10.1016/j.apenergy.2018.06.132.

Csereklyei, Z., Anantharama, N., Kallies, A., 2021. Electricity market transitions in Australia: evidence using model-based clustering. Energy Econ. 103, 105590 https://doi.org/10.1016/j.eneco.2021.105590.

Dluhopolskyi, O., Kozlovskyi, S., Popovskyi, Y., Lutkovska, S., Butenko, V., Popovskyi, T., Mazur, H., Kozlovskyi, A., 2023. Formation of the model of sustainable economic development of renewable energy. Economics 11 (2), 51–78. https://doi.org/10.2478/eoik-2023-0050.

Dogan, E., Seker, F., 2016. Determinants of CO_2 emissions in the European Union: the role of renewable and non-renewable energy. Renew. Energy 94, 429–439. https://doi.org/10.1016/j.renene.2016.03.078.

Dupont, C., Moore, B., Boasson, E.L., Gravey, V., Jordan, A., Kivimaa, P., Kulovesi, K., Kuzemko, C., Oberthür, S., Panchuk, D., Rosamond, J., Torney, D., Tosun, J., Von Homeyer, I., 2024. Three decades of EU climate policy: Racing toward climate neutrality? WIREs Clim. Change 15 (1), e863. https://doi.org/10.1002/wcc.863.

Enerdata, 2022. Global Energy & CO2 Data. ⟨https://www.enerdata.net/research/energy-market-data-co2-emissions-database.html⟩.

EnerData, 2023a. Energy transition. Shape your energy and climate policies. ⟨https://www.enerdata.net/decarbonisation/energy-transition.html⟩ (accessed 21 October 2023).

EnerData, 2023b. Global Energy Transition Statistics. ⟨https://yearbook.enerdata.net/⟩ (accessed 21 October 2023).

EnerData, 2023c. World Energy & Climate Statistics – Yearbook 2023. ⟨https://yearbook.enerdata.net/total-energy/world-consumption-statistics.html⟩.

European Scientific Advisory Board on Climate Change, 2023. Scientific advice for the determination of an EU-wide 2040 climate target and a greenhouse gas budget for 2030–2050. ⟨https://climate-advisory-board.europa.eu/reports-and-publications/scientific-advice-for-the-determination-of-an-eu-wide-2040/scientific-advice-for-the-determination-of-an-eu-wide-2040-climate-target-and-a-greenhouse-gas-budget-for-2030-2050.pdf/⟩@@display-file/file (accessed 21 October 2023).

Food and Agriculture Organization of the United Nations, 2020. Intended Nationally Determined Contribution (INDC) of Ukraine to a New Global Climate Agreement. ⟨http://extwprlegs1.fao.org/docs/pdf/ukr190395.pdf⟩ (accessed 21 October 2023).

Gong, B., Zheng, X., Guo, Q., Ordieres-Meré, J., 2019. Discovering the patterns of energy consumption, GDP, and CO_2 emissions in China using the cluster method. Energy 166, 1149–1167. https://doi.org/10.1016/j.energy.2018.10.143.

Gyimah, J., Nwigwe, U.A., Safi, D.R., Opoku, E.O., Yao, X., 2023. Achieving carbon neutrality in West Africa: the impact of financial development and good governance. PLoS ONE 18 (10), e0293235. https://doi.org/10.1371/journal.pone.0293235.

Harkanth, S., Phulpagar, B., 2013. A survey on clustering methods and algorithms. Int. J. Comput. Sci. Inf. Technol. 4 (5), 687–691. ⟨https://ijcsit.com/docs/Volume%204/Vol4Issue5/ijcsit2013040511.pdf⟩.

He, Z., Xu, S., Shen, W., Long, R., Chen, H., 2017. Impact of urbanization on energy related CO_2 emission at different development levels: regional difference in China based on panel estimation. J. Clean. Prod. 140 (3)), 1719–1730. https://doi.org/10.1016/j.jclepro.2016.08.155.

Hsieh, N., 2005. Hybrid mining approach in the design of credit scoring models. Expert Syst. Appl. 28 (4), 655–665. https://doi.org/10.1016/j.eswa.2004.12.022.

Inekwe, J., Maharaj, E.A., Bhattacharya, M., 2020. Drivers of carbon dioxide emissions: an empirical investigation using hierarchical and non-hierarchical clustering methods. Environ. Ecol. Stat. 27, 1–40. https://doi.org/10.1007/s10651-019-00433-4.

Jaoua, N., 2019. Mathematical Model for CO2 Emissions Reduction to Slow and Reverse Global Warming. In: Tiefenbacher, J.P. (Ed.), Global Warming and Climate Change. IntechOpen, Rijeka. https://doi.org/10.5772/intechopen.88961.

Khvostina, I., Oliinyk, V., Semerikov, S., Solovieva, V., Yatsenko, V., Kohut-Ferens, O., 2021. Hazards and risks in assessing the impact of oil and gas companies on the environment. IOP Conf. Ser.: Earth Environ. Sci. 628 (1), 012027 https://doi.org/10.1088/1755-1315/628/1/012027.

Kim, D.-H., Suen, Y.-B., Lin, S.-C., 2019. Carbon dioxide emissions and trade: evidence from disaggregate trade data. Energy Econ. 78, 13–28. https://doi.org/10.1016/j.eneco.2018.08.019.

Kohonen, T., 2001. Springer Series in Information Sciences: Vol. 30. Self-organizing maps. Springer-Verlag, Berlin. https://doi.org/10.1007/978-3-642-56927-2.

Li, D.-C., Chang, C.-J., Chen, C.-C., Chen, W.-C., 2012. Forecasting short-term electricity consumption using the adaptive grey-based approach — an Asian case. Omega 40 (6), 767–773. https://doi.org/10.1016/j.omega.2011.07.007.

Li, T., Wang, Y., Zhao, D., 2016. Environmental Kuznets curve in China: new evidence from dynamic panel analysis. Energy Policy 91, 138–147. https://doi.org/10.1016/j.enpol.2016.01.002.

Matviychuk, A., Lukianenko, O., Miroshnychenko, I., 2019. Neuro-fuzzy model of country's investment potential assessment. Fuzzy Econ. Rev. 24 (2), 65–88. https://doi.org/10.25102/fer.2019.02.04.

Ministry of Ecology and Natural Resources of Ukraine, 2021. Analytical review of the updated nationally determined contribution of Ukraine to the Paris agreement. ⟨https://mepr.gov.ua/files/docs/klimatychna_polityka/Analytical%20Review%20Updated%20Nationally%20Determined%20Contribution%20of%20Ukraine%20to%20the%20Paris%20Agreement%20July%202021%20⟩(2).pdf (accessed 21 October 2023).

Miroshnychenko, I., Kravchenko, T, Drobyna, Y., 2021. Forecasting electricity generation from renewable sources in developing countries (on the example of Ukraine). Neuro-Fuzzy Model. Tech. Econ. (10), 164–198. https://doi.org/10.33111/nfmte.2021.164.

Mrabet, Z., AlSamara, M., Hezam Jarallah, S., 2017. The impact of economic development on environmental degradation in Qatar. Environ. Ecol. Stat. 24 (1), 7–38. https://doi.org/10.1007/s10651-016-0359-6.

Net zero scorecard, 2023. Energy & Climate Intelligence Unit. ⟨https://eciu.net/netzerotracker⟩ (accessed 21 October 2023).

Oliskevych, M., Beregova, G., Tokarchuk, V., 2018. Fuel consumption in Ukraine: evidence from vector error correction model. Int. J. Energy Econ. Policy 8 (5), 58–63. ⟨https://www.econjournals.com/index.php/ijeep/article/view/6825⟩.

Publications Office of the European Union, 2019. A vision for the European industry until 2030: final report of the Industry 2030 high level industrial roundtable. European Commission. https://doi.org/10.2873/102179 (accessed 21 October 2023).

Relva, S.G., da Silva, V.O., Gimenes, A.L.V., Udaeta, M.E.M., Ashworth, P., Peyerl, D., 2021. Enhancing developing countries' transition to a low-carbon electricity sector. Energy 220, 119659. https://doi.org/10.1016/j.energy.2020.119659.

Sarkodie, S.A., Owusu, P.A., 2017. The relationship between carbon dioxide, crop and food production index in Ghana: By estimating the long-run elasticities and variance decomposition. Environ. Eng. Res. 22 (2), 193–202. https://doi.org/10.4491/eer.2016.135.

Schandl, H., Hatfield-Dodds, S., Wiedmann, T., Geschke, A., Cai, Y., West, J., Newth, D., Baynes, T., Lenzen, M., Owen, A., 2016. Decoupling global environmental pressure and economic growth: scenarios for energy use, materials use and carbon emissions. J. Clean. Prod. 132, 45–56. https://doi.org/10.1016/j.jclepro.2015.06.100.

The World Bank, 2022. World Development Indicators. ⟨https://datacatalog.worldbank.org/search/dataset/0037712⟩.

Turlakova, S., 2022. Modeling the values of reflexive characteristics of agents within the management of herd behavior at the enterprises. Neuro-Fuzzy Model. Tech. Econ. (11), 48–77. https://doi.org/10.33111/nfmte.2022.048.

Turlakova, S., Lohvinenko, B., 2023. Artificial intelligence tools for managing the behavior of economic agents at micro level. Neuro-Fuzzy Model. Tech. Econ. (12), 3–39. https://doi.org/10.33111/nfmte.2023.003.

UN Department of Economic and Social Affairs, 2023. Transforming our world: the 2030 Agenda for Sustainable Development. ⟨https://sdgs.un.org/2030agenda⟩ (accessed 21 October 2023).

UNDP, 2023. What are NDCs and how do they drive climate action? ⟨https://climatepromise.undp.org/news-and-stories/NDCs-nationally-determined-contributions-climate-change-what-you-need-to-know⟩ (accessed 21 October 2023).

UNICEF, 2021. Climate Landscape Analysis for Children (CLAC) in Ukraine. ⟨https://www.unicef.org/ukraine/media/15766/file/Climat%20Landscape%20Analysis%20for%20Children%20⟩(CLAC)%20in%20Ukraine.pdf (accessed 21 October 2023).
Véliz, K.D., Silva, S., Hernández, B., 2023. Vision for a sustainable energy transition and decarbonization: a case study of students surveyed at a Chilean University. Heliyon 9 (11), e21534. https://doi.org/10.1016/j.heliyon.2023.e21534.
Wehrens, R., Kruisselbrink, J., 2018. Flexible self-organizing maps in kohonen 3.0. J. Stat. Softw. 87 (7), 1–18. https://doi.org/10.18637/jss.v087.i07.
Wu, T.-H., Chen, Y.-S., Shang, W., Wu, J.-T., 2018. Measuring energy use and CO_2 emission performances for APEC economies. J. Clean. Prod. 183, 590–601. https://doi.org/10.1016/j.jclepro.2018.02.028.
Xu, D., Tian, Y., 2015. A comprehensive survey of clustering algorithms. Ann. Data Sci. 2, 165–193. https://doi.org/10.1007/s40745-015-0040-1.
Yan, R., Ma, M., Zhou, N., Feng, W., Xiang, X., Mao, C., 2023. Towards COP27: decarbonization patterns of residential building in China and India. Appl. Energy 352, 122003. https://doi.org/10.1016/j.apenergy.2023.122003.
Zhang, S., Zhou, N., Feng, W., Ma, M., Xiang, X., You, K., 2023. Pathway for decarbonizing residential building operations in the US and China beyond the mid-century. Appl. Energy 342, 121164. https://doi.org/10.1016/j.apenergy.2023.121164.
Zharova, L., Zhytkevych, O., Chaplynska, N., 2023. Ukrainian Firms' Transition toward a Low-Carbon Economy: How to Be Proactive. Int. J. Energy, Environ. Econ. 30 (3), 365–384. ⟨https://novapublishers.com/shop/ukrainian-firms-transition-toward-a-low-carbon-economy-how-to-be-proactive/⟩.
Zhou, W.-Y., Yang, W.-L., Wan, W.-X., Zhang, J., Zhou, W., Yang, H.-S., Yang, H., Xiao, H., Deng, S.-H., Shen, F., Wang, Y.-J., 2018. The influences of industrial gross domestic product, urbanization rate, environmental investment, and coal consumption on industrial air pollutant emission in China. Environ. Ecol. Stat. 25 (4), 429–442. https://doi.org/10.1007/s10651-018-0412-8.
Zhytkevych, O., Brochado, A., 2022. Modeling national decarbonization capabilities using Kohonen maps. Neuro-Fuzzy Model. Tech. Econ. (11), 3–24. https://doi.org/10.33111/nfmte.2022.003.
Zhytkevych, O., Matviychuk, A., Kmytiuk, T., 2023. Modeling Nations' Decarbonisation Potential. In: Ortiz-Rodríguez, F., Tiwari, S., Usoro Usip, P., Palma, R. (Eds.), Communications in Computer and Information Science: Vol. 1888. Electronic Governance with Emerging Technologies. Springer, Cham, pp. 60–77. https://doi.org/10.1007/978-3-031-43940-7_6.
Zi, C., Jie, W., Hong-Bo, C., 2016. CO_2 emissions and urbanization correlation in China based on threshold analysis. Ecol. Indic. 61 (2)), 193–201. https://doi.org/10.1016/j.ecolind.2015.09.013.
Zomchak, L., Nehrey, M., Oliskevych, M., Voronenko, I., Rogoza, N., 2023. Economic growth and environmental degradation: data intelligence for sustainable environment. J. Inf. Technol. Manag. 15 (1), 163–177. https://doi.org/10.22059/jitm.2023.91150.